

SURYA

Senior Python Developer

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PROFESSIONAL SUMMARY

Results-driven Senior Python Developer with 10+ years of experience designing, building, and deploying high-performance data pipelines, backend services, and AI/ML systems. Proven expertise across AWS, GCP, and Azure cloud platforms, with deep knowledge of Python ecosystems including Pandas, NumPy, PySpark, LangChain, DSPy, and MLflow. Extensive experience in Big Data engineering, LLM-driven application development, agentic AI workflows, and real-time streaming architectures. Strong background delivering production-grade solutions in regulated industries including healthcare (HIPAA), finance (PCI-DSS/SOX), and aviation (FAA/IATA).

TECHNICAL SKILLS

Programming Languages: Python (Expert), PySpark, Scala, Java, SQL, T-SQL, Unix/Bash, Spring Boot

Python Libraries / ML: NumPy, Pandas, Scikit-Learn, Matplotlib, MLlib, DSPy, LangChain, LangGraph, MLflow, Gradient Boosting, Random Forest, Logistic Regression, Decision Tree

Cloud Platforms: AWS (S3, Glue, EMR, Lambda, ECS, EKS, EC2, Sagemaker, Redshift, RDS, EventBridge, CloudWatch, CDK, SQS, SNS, Athena), GCP (BigQuery, DataProc, GKE, Spanner, Functions, Cloud Storage), Azure (Data Factory, Databricks, SQL, Data Lake, Blob Storage, HDInsight, CosmosDB)

Big Data Ecosystem: Apache Spark, Kafka, Airflow, Dagster, Hive, HDFS, NiFi, Spark Streaming, Zookeeper, Sqoop, Flume, MapReduce, YARN

Databases & Tools: MySQL, PostgreSQL, MongoDB, Oracle 11g/10g, MS SQL Server, Teradata, Amazon Redshift, Snowflake, CosmosDB

ETL & Integration: AWS Glue, Informatica PowerCenter 10.x/9.6, Talend 5.6, SSIS, Semarchy xDM

DevOps & Containerization: Docker, Kubernetes, Jenkins, Git, AWS CDK, Terraform, AWS WAF

Reporting & BI: Tableau, Power BI, Grafana, Prometheus, Splunk, Collibra

Methodologies: Agile/Scrum, Waterfall, SDLC, CI/CD, DataOps, MLOps

PROFESSIONAL EXPERIENCE

Humana: August 2023 – Present (Location: Louisville, KY)

Role: Sr. Python / Big Data Engineer

Responsibilities:

- Designed and deployed agentic retrieval workflows using **Python**, **LangChain**, and **LangGraph** to build context-aware AI agents ingesting PR metadata, commit metadata, and GitHub history, enabling automated CES scoring and code quality metric computation for healthcare pipelines while ensuring HIPAA-compliant audit trails and HL7 interoperability across AWS S3, Lambda, and Glue.
- Built production-grade LLM-driven scoring systems using **DSPy**, integrating prompt optimization, chaining, and context packing to automate healthcare PR-level scoring, leveraging GPT-4, Claude, AWS Sagemaker, and **MLflow** for model versioning, eval set generation, and feedback loops under HIPAA-compliant governance.
- Migrated legacy orchestration from **Apache Airflow** to **Dagster** across 40+ production pipelines processing healthcare claims, clinical records, and patient data on AWS ECS and EKS, implementing retries, backfills, scheduling, and data quality alerts with Power BI and Collibra reporting frameworks.
- Engineered backend services using **Python**, **AWS Lambda**, EventBridge, and AWS CDK to ingest PR and commit metadata from GitHub and Jenkins, linking artifacts to compute CES metrics, quality signals, and anomalies while exposing results through RESTful APIs and Power BI dashboards under HIPAA compliance.
- Developed data modeling frameworks using **Pandas**, **NumPy**, and **MLlib** to link PRs, commits, branches, and epics for healthcare software delivery metrics, building graph-based relationships across Jira, GitHub, and AWS S3, implementing anomaly detection, role classification, and effort estimation for HIPAA-compliant reporting.
- Orchestrated large-scale backfills and historical corrections using **Dagster** and **AWS Glue**, processing over 10 million commits and clinical record deltas with batching, caching, and DSPy-based score optimization, ensuring data quality alerts captured in AWS CloudWatch and Collibra lineage tracking.
- Implemented **SonarQube** integration for code-quality scoring, linking static analysis results with GitHub PR metadata and commit history to enrich CES metrics, while building Jira-integrated workflows for epic/work-type classification and automated tagging under HIPAA-compliant metadata governance.

- Developed and deployed AI/ML pipelines using **MLlib** and **NumPy** to predict patient readmission risk, orchestrated with Airflow for ingestion, transformation, and model retraining, ensuring HIPAA-compliant data handling and HL7 interoperability with comprehensive audit trails in alignment with HITECH Act regulations.
- Built observability and monitoring solutions using **AWS CloudWatch**, Lambda, and Dagster asset sensors to track metrics, traces, logs, and data quality alerts across healthcare ETL pipelines, enabling automated alerting for failures, anomaly detection, and SLA breaches with Collibra and ERWIN lineage tracking.
- Engineered AI/ML clinical decision support systems using **LLMs** and **MLlib**, processing unstructured EHR data, identifying high-risk patients, and deploying inference infrastructure on AWS EC2 with encryption and role-based access to meet HIPAA and HITECH security requirements for real-time healthcare workflows.

Environment: Python, LangChain, LangGraph, DSPy, MLflow, MLlib, Pandas, NumPy, Dagster, Apache Airflow, AWS Sagemaker, AWS Lambda, AWS ECS, AWS EKS, AWS Glue, AWS S3, AWS CDK, AWS CloudWatch, AWS EventBridge, AWS EC2, AWS RDS, GitHub, Jenkins, SonarQube, Power BI, Collibra, Jira, ERWIN, Git

Morgan Stanley: October 2020 – July 2023 (Location: New York, NY)

Role: Sr. Python / Data Engineer

Responsibilities:

- Designed and deployed production-grade **Python** data pipelines on AWS, ingesting PR metadata, commit telemetry, and code quality signals from GitHub and Jenkins, integrating AWS Lambda, S3, Glue, and ECS to ensure PCI-DSS-compliant financial workflows with automated retries, backfills, and full observability.
- Built retrieval agents using **Python** and **DSPy** to extract historical code context, diffs, and commit histories from GitHub, implementing prompt chaining and context packing for LLM optimization, automating score correction workflows, and exposing CES metrics via REST APIs with SOX and GLBA-compliant secure audit trails.
- Migrated orchestration from **Apache Airflow** to **Dagster**, enhancing PR/commit linking, branch tracking, and epic linking across Jira, GitHub, and banking systems, integrating AWS EventBridge, SQS, and Lambda while maintaining PCI-DSS and SOX compliance with Power BI dashboard exposure.
- Developed anomaly detection and data quality monitoring using **Python**, **AWS Glue**, and **AWS CloudWatch**, identifying deviations in commit metadata and code quality signals from SonarQube and PMD, triggering automated alerts and implementing DSPy-based score correction under GLBA-compliant audit workflows.
- Implemented observability frameworks using **AWS CloudWatch**, **AWS X-Ray**, and Python scripts, tracking metrics, traces, and logs across pipelines with automated notifications through SQS and Slack, ensuring PCI-DSS, SOX, and GLBA compliance while maintaining continuous financial data reliability.
- Designed and executed **DSPy**-based eval sets and feedback loops, collecting user feedback on scoring accuracy, retraining LLM/prompt optimization models stored in AWS S3 and RDS, implementing batching and caching to optimize performance and cost under GLBA and SOX-compliant secure financial workflows.
- Engineered AWS-native pipelines using ECS, Lambda, S3, RDS, and CDK to automate backfills, retries, and scheduling for high-volume financial data ingestion, integrating GitHub webhooks and Jenkins CI/CD pipelines with PCI-DSS encryption and IAM access controls for secure cardholder data processing.
- Integrated **SonarQube** and **PMD** results with Python backend services to compute CES and quality metrics, linking findings to PR metadata, commit metadata, and Jira epics, deploying Lambda functions triggered by EventBridge with results stored in S3 and RDS under PCI-DSS, SOX, and GLBA compliance.
- Implemented **AWS WAF** rules and security policies for banking applications, mitigating SQL injection, XSS, and DDoS attacks, integrating CloudWatch monitoring, geo-blocking, and rate-limiting configurations, ensuring GLBA compliance and secure customer authentication workflows across digital banking platforms.
- Developed infrastructure-as-code templates using **AWS CDK** in Python/TypeScript to provision EC2, VPCs, security groups, IAM roles, and backup policies for loan origination systems, automating deployments, enforcing least-privilege access, and ensuring PCI-DSS and SOX compliance with version-controlled pipelines.

Environment: Python, DSPy, Dagster, Apache Airflow, AWS Lambda, AWS ECS, AWS EKS, AWS S3, AWS RDS, AWS Glue, AWS CloudWatch, AWS X-Ray, AWS CDK, AWS EventBridge, AWS SQS, AWS WAF, GitHub, Jenkins, SonarQube, PMD, Power BI, Jira, ERWIN, E/R Studio, Git, JWT

Delta Airlines: December 2017 – September 2020 (Location: Atlanta, GA)

Role: Python Data Engineer / Data Analyst

Responsibilities:

- Designed and deployed production-grade data pipelines using **Python** and **Apache Airflow** to ingest commit metadata and PR metadata from GitHub, integrating AWS Lambda and EventBridge for real-time processing, exposing quality metrics via REST APIs, and ensuring SOC 2 audit trails and FAA/IATA compliance.

- Built end-to-end data modeling frameworks using **Python** to link PRs and commits, tracking branch merges to main repositories, integrating AWS Glue and **Dagster** pipelines for historical backfills, and implementing Prometheus and Grafana dashboards for anomaly detection and IOSA-compliant aviation quality monitoring.
- Engineered retrieval agents using **Python** to extract contextual metadata from GitHub, integrating **SonarQube** outputs into AWS S3 data lakes, computing technical debt and maintainability metrics, and exposing dashboards via Power BI with observability through Prometheus for continuous FAA-compliant quality improvement.
- Developed analytics platforms using **Python**, AWS RDS, and **Power BI** to generate CES metrics, correlating PR and commit metadata with Jira epics, designing schemas in ERWIN and E/R Studio to support graph-like linking and epic-level reporting aligned with FAA documentation and safety-critical software audit readiness.
- Built observability frameworks using **Prometheus**, **Grafana**, and AWS CloudWatch, tracking Airflow DAG execution times, AWS Glue job success rates, and Lambda invocation patterns, implementing Python scripts and S3 pipelines for audit trail logging, enabling SOC 2 compliance and root-cause analysis.
- Integrated **SonarQube** with Airflow and AWS Glue, ingesting static analysis metrics into AWS S3, mapping code severity to FAA DAL levels, implementing **Python** logic to generate automated risk scoring, and visualizing defect densities and technical debt via Power BI for aviation software compliance.
- Designed backend services using **Python** on AWS Lambda and EC2, processing thousands of daily PRs and commits, integrating Jenkins CI/CD pipelines with PMD static analysis, storing data in AWS S3, and ensuring IOSA-compliant traceability, FAA version control governance, and safety-critical code quality monitoring.
- Implemented distributed caching architecture using **AWS ElastiCache**, optimizing frequently accessed aircraft maintenance records, integrating PMD static analysis in CI/CD pipelines for Java-based aviation systems, ensuring FAA maintenance approval compliance, and improving response times for pre-flight inspections.

Environment: Python, AWS Lambda, AWS Glue, AWS ECS, AWS EKS, AWS EC2, AWS RDS, AWS S3, AWS CDK, AWS SQS, AWS SNS, AWS WAF, Apache Airflow, Dagster, DSPy, SonarQube, PMD, Prometheus, Grafana, Power BI, Collibra, GitHub, Jenkins, ERWIN, E/R Studio, Jira, Git

Cisco: February 2016 – November 2017 (Location: San Jose, CA)

Role: Python Data Engineer / Migration Specialist

Responsibilities:

- Built a Network Data Integration platform using Azure Cloud and **Python**, improving enterprise data integration efficiency and enabling seamless data flow across global networking systems by consolidating monitoring telemetry from thousands of devices deployed in international data centers.
- Developed scalable data ingestion pipelines using **Python**, **Talend ETL**, and **Azure Data Factory**, efficiently processing large volumes of network monitoring and security appliance data while ensuring reliable ingestion, transformation, and availability for downstream operational analytics.
- Executed data migration initiatives transferring configuration data from legacy network management systems to modern **SDN** platforms, maintaining configuration accuracy and uninterrupted network services throughout the transition to software-defined infrastructure.
- Built complex orchestration workflows using **Python** and **Apache Airflow**, coordinating ETL processes across distributed network monitoring systems in multiple global regions while aligning execution schedules with regional compliance and operational time zone requirements.
- Developed predictive capacity forecasting models using **Python** and **Azure Databricks**, analyzing historical traffic trends to improve bandwidth planning accuracy and support proactive decision making for global network growth requirements.
- Created real-time network anomaly detection solutions using **Python** streaming analytics, enabling early identification of security threats and performance degradation by processing high-volume network event streams continuously.
- Implemented data security and privacy controls aligned with international regulations, applying **Python**-based encryption, role-based access controls, and data anonymization techniques while handling sensitive network configuration and traffic analysis datasets.
- Designed and implemented data quality frameworks using **Python** for validating global network configuration data, ensuring consistency and correctness of configuration changes prior to deployment across distributed infrastructure environments.

Environment: Python, Azure Cloud, Azure Data Factory, Azure SQL Database, Azure Data Lake Storage, Azure Databricks, Talend ETL, Apache Airflow, Streaming Analytics, SDN Platforms, Git

Flipkart: August 2014 – December 2015 (Location: Bangalore, India)

Role: Python ETL Developer

Responsibilities:

- Built a scalable ETL pipeline using **Python**, AWS Services, Informatica, and Talend, integrating data from e-commerce platforms, inventory systems, and customer databases to create a unified data flow supporting high-volume daily transaction processing across multiple online retail channels.
- Configured Amazon S3 for secure storage of raw, staging, and transformed product catalog datasets, enabling reliable **Python**-driven ingestion workflows for large SKU volumes while applying versioning and lifecycle policies to balance accessibility with long-term storage cost optimization.
- Utilized **Python** and **Informatica PowerCenter** to extract data from MySQL, PostgreSQL, and MongoDB sources containing customer profiles and purchase history, ensuring accurate extraction, preserved relationships, and consistent processing of high-volume customer records.
- Developed optimized **Python** and **Talend** jobs processing CSV and JSON-based vendor feeds and marketplace listings, improving data processing efficiency and supporting timely product catalog updates across web and mobile commerce platforms.
- Designed AWS Glue jobs using **Python** to catalog, transform, and load curated sales and inventory data into Amazon Redshift, optimizing storage layouts and enabling efficient analytical queries supporting sales performance and category-level insights.
- Automated ETL job scheduling using **Python**, **AWS Lambda**, and Amazon CloudWatch Events, improving reliability of time-sensitive processes such as daily sales reporting and inventory refreshes through managed execution and dependency handling.
- Connected Tableau to Amazon Redshift and implemented **Python**-driven data preparation workflows to deliver interactive dashboards for real-time analysis of sales trends, inventory positions, and customer behavior, enabling data-driven merchandising decisions.

Environment: Python, AWS S3, AWS Glue, AWS Lambda, AWS CloudWatch, Amazon Redshift, Informatica PowerCenter, Talend, Talend Open Studio, MySQL, PostgreSQL, MongoDB, Tableau, Git

EDUCATION

Bachelor of Technology in Computer Science & Engineering — Koneru Lakshmaiah University, 2010–2014

CERTIFICATIONS & KEY COMPETENCIES

- AWS Certified Data Analytics / Solutions Architect (or equivalent cloud certification)
- 10+ years of Python development across data engineering, backend services, AI/ML, and LLM applications
- Domain expertise: Healthcare (HIPAA/HITECH/HL7), Financial Services (PCI-DSS/SOX/GLBA), Aviation (FAA/IATA/SOC 2)
- Strong foundation in agentic AI, RAG systems, LLM prompt engineering, and production MLOps workflows